Exercise - 3.3

1. Find the values of (i) $\sin(480^\circ)$ (ii) $\sin(-1110^\circ)$ (iii) $\cos(300^\circ)$ (iv) $\tan(1050^\circ)$ (v) $\cot(660^\circ)$ (vi) $\tan\left(\frac{19\pi}{3}\right)$ (vii) $\sin\left(-\frac{11\pi}{3}\right)$.
2. $\left(\frac{5}{7}, \frac{2\sqrt{6}}{7}\right)$ is a point on the terminal side of an angle θ in standard position. Determine the six trigonometric function values of angle θ.
3. Find the values of other five trigonometric functions for the following:
   (i) $\cos\theta = -\frac{1}{2}$, θ lies in the III quadrant.
   (ii) $\cos\theta = \frac{2}{3}$, θ lies in the I quadrant.
   (iii) $\sin\theta = -\frac{2}{3}$, θ lies in the IV quadrant.
   (iv) $\tan\theta = -2$, θ lies in the II quadrant.
   (v) $\sec\theta = \frac{13}{5}$, θ lies in the IV quadrant.
4. Prove that $\frac{\cot(180^\circ+\theta)\sin(90^\circ-\theta)\cos(-\theta)}{\sin(270^\circ+\theta)\tan(-\theta)\text{cosec}(360^\circ+\theta)} = \cos^2\theta\cot\theta$.
5. Find all the angles between 0° and 360° which satisfy the equation $\sin^2\theta = \frac{3}{4}$.
6. Show that $\sin^2\frac{\pi}{18} + \sin^2\frac{\pi}{9} + \sin^2\frac{7\pi}{18} + \sin^2\frac{4\pi}{9} = 2$.

3.5 Trigonometric Identities

3.5.1 Sum and difference identities or compound angles formulas

Now, compound angles are algebraic sum of two or more angles. Trigonometric functions do not satisfy the functional relations like $f(x+y) = f(x) + f(y)$ and $f(kx) = kf(x)$, k is a real number. For example, $\cos(\alpha+\beta) \neq \cos\alpha + \cos\beta$, $\sin(2\alpha) \neq 2\sin\alpha$, $\tan 3\alpha \neq 3\tan\alpha, \ldots$. Thus, we need to derive formulas for $\sin(\alpha+\beta), \cos(\alpha+\beta), \ldots$ and use them in calculations of application problems.

Music is made up of vibrations that create pressure on ear-drums. Musical tones can be modeled with sinusoidal graphs (graphs looks like that of $y = \sin x$ or $y = \cos x$). When more than one tone is played, the resulting pressure is equal to the sum of the individual pressures. In this context sum and difference trigonometric identities are used as an important application. Also, sum and difference trigonometric identities are helpful in the analysis of waves.

First we shall prove the identity for the cosine of the sum of two angles and extend it to prove all other sum or difference identities.

Identity 3.1: $\cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$

Proof. Consider the unit circle with centre at O. Let $P = P(1,0)$.

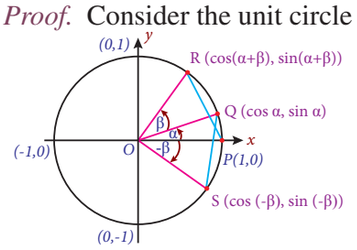


Figure 3.15

Let Q, R and S be points on the unit circle such that $\angle POQ = \alpha$, $\angle POR = \alpha+\beta$ and $\angle POS = -\beta$ as shown in the Figure 3.15. Clearly, angles $\alpha, \alpha+\beta$ and $-\beta$ are in standard positions. Now, the points Q, R and S are given by $Q(\cos\alpha, \sin\alpha)$, $R(\cos(\alpha+\beta), \sin(\alpha+\beta))$ and $S(\cos(-\beta), \sin(-\beta))$.